Platonic Solids

An attempt

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Introduction to the 5 Platonic solids

Platonic solids are a special class of highly symmetric three-dimensional geometric objects. They are named after the Greek philosopher Plato, who discussed them in his work *Timaeus* and associated them with the classical elements of nature. However, the mathematical study of these solids predates Plato and was known to earlier Greek mathematicians.

A Platonic solid is a **regular convex polyhedron** satisfying the following conditions:

- All faces are congruent regular polygons.
- The same number of faces meet at every vertex.
- All edges are equal in length.
- All vertices are geometrically identical.

Because of these strict conditions, only **five** Platonic solids exist.

Platonic Solid	Faces	Face Type	Vertices	Edges	Hamiltonian	Eulerian
Tetrahedron	4	Equilateral Triangle	4	6	Yes	No
Cube (Hexahedron)	6	Square	8	12	Yes	No
Octahedron	8	Equilateral Triangle	6	12	Yes	Yes
Dodecahedron	12	Regular Pentagon	20	30	Yes	No

Icosahedron	20	Equilateral Triangle	12	30	Yes	No

The existence of only five Platonic solids can be understood through geometric constraints. At each vertex, the sum of the face angles meeting there must be **less than** (360°); otherwise, the shape would lie flat and fail to form a three-dimensional solid. This restriction allows only a limited number of regular polygons to combine in space, resulting in exactly five possible solids.

The Platonic solids possess remarkable symmetry and play an important role in geometry, topology, crystallography, architecture, chemistry, and mathematical modelling. Several important mathematical concepts such as duality, symmetry groups, Euler's formula, and polyhedral geometry can be explored through their study.

These solids are closely related through the concept of **duality**. The tetrahedron is self-dual, while the cube and octahedron form a dual pair, as do the dodecahedron and icosahedron.

The study of Platonic solids provides a foundation for understanding regular polyhedra and offers insight into the deep relationship between symmetry, geometry, and spatial structure.

Historical Perspective

The Platonic solids occupy a unique place in the history of mathematics. They are the only five convex polyhedra in which every face is the same regular polygon and the same number of faces meet at every vertex. Beyond their geometric definition, they have played a profound role in the development of mathematical thought, philosophy, and early attempts to understand the structure of the physical world.

What makes these shapes historically significant is not merely their mathematical rarity, but the way ancient thinkers interpreted them. To the Greek philosophers, geometry was not an abstract discipline detached from reality; it was the language through which the universe expressed order and harmony. The Platonic solids were therefore seen not just as shapes, but as fundamental building blocks of nature itself.

Early Geometric Thought and the Pythagorean Influence

The origins of interest in regular solids can be traced back to the Pythagorean school around the 5th century BCE. The Pythagoreans believed that numbers and geometric relationships governed the structure of reality. Their studies were deeply intertwined with mysticism and philosophy, and they considered geometric perfection to reflect cosmic order.

During this period, certain regular solids such as the cube and tetrahedron were already known. However, they were not yet understood as part of a complete classification. Instead, they were treated as isolated objects with aesthetic and symbolic significance.

The Pythagorean worldview set the stage for a deeper investigation: if symmetry and regularity are fundamental principles of nature, then how many perfectly symmetric three-dimensional forms can exist?

Theaetetus and the Classification of the Five Solids

A major turning point in the history of Platonic solids came with the work of Theaetetus, a mathematician associated with Plato's Academy in the 4th century BCE. He is credited with the first systematic study and classification of all convex regular polyhedra.

The key mathematical achievement attributed to Theaetetus is the proof that there are exactly five such solids. This is not an obvious fact; it arises from strict geometric constraints. A regular polyhedron must have: - identical regular polygonal faces, - identical vertices, - and enough angular deficit at each vertex to allow the shape to close in three dimensions.

These constraints severely limit the possibilities. The result is striking: only five configurations satisfy all conditions simultaneously. This classification transformed the Platonic solids from a set of known shapes into a complete mathematical system.

Plato's Philosophical Interpretation

Although Plato was not the discoverer of these solids, his philosophical influence is the reason they became historically prominent. In his dialogue *Timaeus*, Plato proposed a cosmological framework in which the physical world is composed of four classical elements: earth, water, air, and fire, along with a fifth element representing the heavens or cosmos.

He associated each of the four elements with a Platonic solid based on perceived geometric and physical properties:

- The tetrahedron, with its sharp edges and pointed structure, was associated with fire.
- The cube, stable and evenly balanced, was associated with earth.
- The octahedron, relatively symmetrical and less "grounded," was associated with air.
- The icosahedron, with its many faces and smooth appearance, was associated with water.
- The dodecahedron, the most complex and least intuitively tangible, was associated with the structure of the universe itself.

These associations were not arbitrary aesthetic choices. They reflect an early attempt to connect geometric structure with physical intuition. Sharpness, stability, fluidity, and symmetry were interpreted as geometric expressions of natural behavior.

Euclid and the Mathematical Formalization

The Platonic solids were later rigorously treated by Euclid in *Elements*, one of the most influential mathematical texts in history. Euclid provided formal geometric constructions and proofs that established these solids within a deductive system.

Where earlier interpretations were philosophical or intuitive, Euclid's approach was strictly logical. He demonstrated how each solid could be constructed from fundamental geometric principles and confirmed that no other regular convex polyhedra exist.

This marked an important transition in the history of mathematics: from geometric intuition rooted in philosophy to a formal axiomatic system.

Kepler and the Platonic Model of Planetary Orbits

The influence of Platonic solids did not end in antiquity. During the Renaissance, the German astronomer Johannes Kepler attempted to revive the idea that geometry lies at the foundation of the physical universe, extending it from elemental theory to planetary motion.

In his early work *Mysterium Cosmographicum* (1596), Kepler proposed a striking model of the solar system in which the spacing between the known planetary orbits was determined by nesting the five Platonic solids inside one another. Each solid was placed between two planetary spheres, with the idea that the ratios of orbital radii could be explained through this geometric arrangement.

In this model, the sequence of solids was used to account for the six known planets at the time (Mercury through Saturn). Although the model was elegant and deeply influenced by the Platonic tradition of mathematical harmony, it was ultimately based on idealized assumptions rather than precise observational data.

Kepler's work is important not because it was correct, but because it represented a transition in scientific thinking. It marked an attempt to move from philosophical geometry—where shapes were assumed to dictate nature—to mathematical physics, where geometric models had to be tested against empirical measurements.

Later in his career, Kepler abandoned the Platonic solid model after analyzing the precise astronomical observations of Tycho Brahe. This led him instead to the discovery that planetary orbits are elliptical, not circular, and governed by three fundamental laws of planetary motion. These laws became a cornerstone of classical mechanics.

Despite being replaced, Kepler's early model remains historically significant. It reflects one of the last major attempts to unify cosmic structure with Platonic geometry, showing how deeply the idea of symmetry influenced early scientific thought.

Why Only Five Solids Exist

The limitation to five Platonic solids arises from a fundamental geometric constraint involving angles at a vertex. For a shape to exist in three dimensions, the sum of the angles meeting at each vertex must be less than 360 degrees; otherwise, the structure would flatten.

Only a small number of regular polygon combinations satisfy this condition. Triangles, squares, and pentagons are the only regular polygons that can form such closed structures in a consistent way. When combined under the requirement of identical vertices and faces, the possibilities reduce precisely to five distinct solids.

This fact highlights a deeper principle: symmetry in higher dimensions is heavily constrained, and only a few perfect structures can exist.

Proof using Shlafli symbol

A Platonic solid is represented by the Schläfli symbol:

$$\{p, q\}$$

where: - p = number of sides of each face
- q = number of faces meeting at each vertex

Each regular p-gon has interior angle:

$$\alpha = \left(1 - \frac{2}{p}\right)\pi$$

At each vertex, q such angles meet. For the solid to exist in 3D space, the sum must be less than 2π:

$$q\left(1 - \frac{2}{p}\right)\pi < 2\pi$$

Divide by π:

$$q\left(1 - \frac{2}{p}\right) < 2$$

Rewriting:

$$q\frac{p-2}{p} < 2$$

Multiply by p:

$$q(p-2) < 2p$$

Rearrange:

$$(p-2)(q-2) < 4$$

We now search integers $p \geq 3$, $q \geq 3$ satisfying:

$$(p-2)(q-2) < 4$$

Checking cases:

- $p = 3 \Rightarrow q = 3, 4, 5$
- $p = 4 \Rightarrow q = 3$
- $p = 5 \Rightarrow q = 3$
- $p \geq 6 \Rightarrow$ no solutions

Thus we obtain exactly five possibilities:

$$\{3,3\}, \{3,4\}, \{4,3\}, \{3,5\}, \{5,3\}$$

Proof using Euler's formula

Let: - V = vertices
- E = edges
- F = faces

Euler's formula:

$$V - E + F = 2$$

Assume: - each face is a regular p-gon
- q faces meet at each vertex

Counting edges:

$$pF = 2E$$

$$qV = 2E$$

So:

$$V = \frac{2E}{q}, \quad F = \frac{2E}{p}$$

Substitute into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2$$

Divide by E:

$$\frac{2}{q} - 1 + \frac{2}{p} = \frac{2}{E}$$

Since $E > 0$, the left side must be positive:

$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$

Test $p, q \geq 3$:

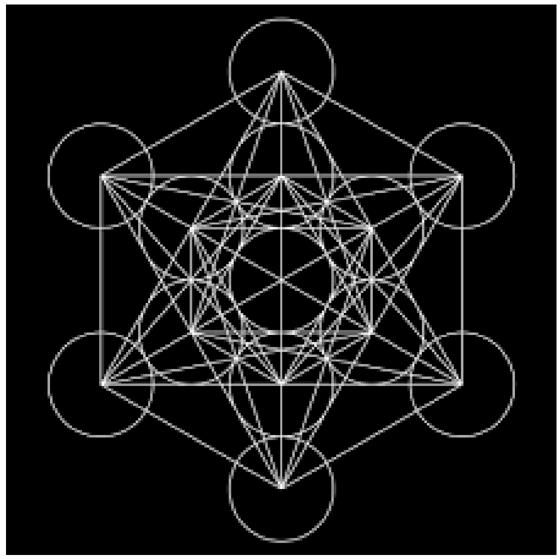
- $p = 3 \Rightarrow q = 3, 4, 5$
- $p = 4 \Rightarrow q = 3$
- $p = 5 \Rightarrow q = 3$
- $p \geq 6 \Rightarrow$ no solutions

Again we obtain:

$$\{3,3\}, \{3,4\}, \{4,3\}, \{3,5\}, \{5,3\}$$

Solid	Schläfli symbol
Tetrahedron	{3,3}
Cube	{4,3}
Octahedron	{3,4}
Dodecahedron	{5,3}
Icosahedron	{3,5}

Metatron's Cube and Its Connection to Platonic Solids



Metatron's Cube is a geometric figure that emerged much later than the classical study of Platonic solids, within traditions of sacred geometry that developed in the medieval and early modern periods. Although it is not part of Greek mathematics, it is deeply influenced by the same idea that symmetry is the foundation of structure in nature.

The construction of Metatron's Cube begins with a pattern of evenly spaced circles arranged in a hexagonal lattice. This arrangement creates a highly regular field of intersection points. When the centers of these circles are identified and connected by straight lines, a complex network of vertices and edges is formed. This resulting structure is what is referred to as Metatron's Cube.

What makes this figure significant is not the circles themselves, but the symmetry they enforce on the resulting network. The arrangement produces a dense system of equal-length connections and repeated angular relationships. Within this system, one can observe multiple embedded configurations that mirror the adjacency patterns of the Platonic solids.

The connection to Platonic solids arises from the fact that these solids are themselves complete symmetry objects in three dimensions. Each Platonic solid corresponds to a distinct symmetry group, and together they exhaust all possible regular convex polyhedra. Metatron's Cube, when viewed as a two-dimensional projection of a highly symmetric structure, encodes these same symmetry constraints in a flattened form. Specific selections of vertices within the figure can reproduce the connectivity patterns of the tetrahedron, cube, octahedron, icosahedron, and dodecahedron.

From this perspective, the relationship is not one of literal containment, but of shared structure. The Platonic solids exist as three-dimensional realizations of symmetry, while Metatron's Cube represents a planar manifestation of a similarly constrained system. Both reflect the same underlying principle: that only a limited number of perfectly regular forms can exist under strict symmetry conditions.

Symbolically, Metatron's Cube has been interpreted as a unifying diagram of geometric order. It is often associated with the idea that all structured forms in nature can be derived from a single underlying pattern of symmetry. In this interpretation, the Platonic solids represent stable three-dimensional outcomes of that pattern, while Metatron's Cube represents the relational framework from which such outcomes can be conceptually reconstructed.

Thus, rather than being a literal container of Platonic solids, Metatron's Cube functions as a symbolic map of symmetry itself, linking simple geometric repetition to the emergence of highly regular spatial forms.

Sacred Geometry

Sacred geometry is not a formal branch of mathematics, but rather a historical and philosophical way of interpreting geometric patterns as expressions of order in nature. At its core, it refers to the belief that certain shapes and proportions—especially those with high symmetry—are fundamentally connected to the structure of the physical world and, in some traditions, to human perception of beauty and harmony.

Although the term "sacred" suggests a mystical origin, the underlying content is deeply mathematical. Sacred geometry is primarily concerned with patterns such as circles, triangles, spirals, and polyhedra, and with the relationships between them. Many of these shapes appear repeatedly in both natural systems and mathematical constructions, which is why they have been historically regarded as meaningful.

From a modern perspective, what sacred geometry attempts to capture can often be explained through symmetry, scaling laws, and optimization principles. For example, the hexagonal pattern seen in honeycombs arises because it efficiently fills space while minimizing material usage. Similarly, spiral patterns appear in galaxies, shells, and plants due to growth processes governed by simple iterative rules.

In this sense, sacred geometry can be interpreted as an early attempt to recognize that nature is not random, but structured by consistent mathematical principles. Before the development of formal physics and biology, geometric forms were one of the few available tools for describing this hidden order.

The human connection to these patterns is also significant. Our visual system is highly sensitive to symmetry, repetition, and proportional relationships. This is why shapes associated with sacred geometry—such as circles, stars, and Platonic solids—often feel aesthetically "balanced" or harmonious. The perception of beauty in these forms is not arbitrary; it is closely linked to how the brain processes structure and regularity.

Thus, sacred geometry can be understood in two complementary ways. Historically, it reflects the attempt to explain nature through ideal forms. Scientifically, it corresponds to the study of symmetry and pattern formation in physical systems. In both cases, it highlights a central idea: that complex structures in nature often arise from simple and repeated geometric rules.

Tetrahedron

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	4
Edges	6
Vertices	4
Face Type	Equilateral Triangle
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(\frac{1}{3}\right) \approx 70.53^\circ$
Surface Area	$\sqrt{3}a^2$
Volume	$\frac{\sqrt{2}}{12}a^3$
Height	$\sqrt{\frac{2}{3}}a$
Inradius (r)	$\frac{\sqrt{6}}{12}a$
Circumradius (R)	$\frac{\sqrt{6}}{4}a$
Midradius (ρ)	$\frac{\sqrt{2}}{4}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$\text{Spec}(A) = \{(3)^1, (-1)^3\}$

Laplacian Matrix

$$L = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

$\mathrm{Spec}(L) = \{(4)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(2)^3, (0)^1\}$

Symmetry and Automorphisms

For Platonic solids, the total number of geometric symmetries of the solid is equal to the total number of automorphisms of its associated graph. In other words, every graph automorphism corresponds to a geometric symmetry, and every geometric symmetry induces a graph automorphism.

$|Sym(Solid)| = |Aut(G)|$

The total number of symmetries is calculated by counting all distinct rotations and reflections about the axes of symmetry passing through vertices, edges, and faces of the solid. These symmetries together form the symmetry group of the Platonic solid.

To count:

Identity = 1

Vertex to centre of opposite face -> 4 axes and each face is a triangle -> 2 rotations per axis $(120°, 240°) = 4 \times 2 = 8$

Edge to opposite edge -> 3 axes and total rotations per axis -> 1 $(180°) = 3 \times 1$ $= 3$

Total rotations = 1+8+3 = 12

$|Rot(Tetrahedron)| = 12$

All rotational symmetries of the tetrahedron correspond to even permutations of the 4 vertices.

$Rot(Tetrahedron) \cong A_4$

Total reflections = Total rotations = 12

Total symmetries = total rotations + total reflections = 12+12 = 24

$|Sym(Tetrahedron)| = 24$

$Sym(Tetrahedron) \cong S_4$

Cube

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	6
Edges	12
Vertices	8
Face Type	Square
Face Interior Angle	$90°$
Dihedral Angle	$90°$
Surface Area	$6a^2$
Volume	a^3
Space Diagonal	$\sqrt{3}a$
Face Diagonal	$\sqrt{2}a$
Inradius (r)	$\frac{a}{2}$
Circumradius (R)	$\frac{\sqrt{3}}{2}a$
Midradius (ρ)	$\frac{\sqrt{2}}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$\mathrm{Spec}(A) = \{(3)^1, (1)^3, (-1)^3, (-3)^1\}$

Laplacian Matrix

$$L = \begin{bmatrix} 3 & -1 & 0 & -1 & -1 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 0 & -1 & 0 \\ -1 & 0 & -1 & 3 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 3 & -1 & 0 & -1 \\ 0 & -1 & 0 & 0 & -1 & 3 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 & 3 & -1 \\ 0 & 0 & 0 & -1 & -1 & 0 & -1 & 3 \end{bmatrix}$$

$\text{Spec}(L) = \{(6)^1, (4)^3, (2)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{6})^1, (2)^3, (\sqrt{2})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 3 axes and each face is a square -> 3 rotations per axis $(90^\circ, 180^\circ, 270^\circ) = 3 \times 3 = 9$

Vertex to opposite vertex -> 4 axes and each vertex has degree 3 -> 2 rotations per axis $(120^\circ, 240^\circ) = 4 \times 2 = 8$

Midpoint of opposite edges -> 6 axes and total rotations per axis -> 1 $(180^\circ) = 6 \times 1 = 6$

Total rotations = 1+9+8+6 = 24

$|Rot(Cube)| = 24$

All rotations of the cube correspond to permutations of the 4 body diagonals.

$Rot(Cube) \cong S_4$

Total reflections = Total rotations = 24

Total symmetries = total rotations + total reflections = 24+24 = 48

$|Sym(Cube)| = 48$

$Sym(Cube) \cong S_4 \times C_2$

Octahedron

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Geometric Properties

Let the edge length be a.

Property	Formula / Value
Faces	8
Edges	12
Vertices	6
Face Type	Equilateral Triangle
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(-\frac{1}{3}\right) \approx 109.47^\circ$
Surface Area	$2\sqrt{3}\,a^2$

Property	Formula / Value
Volume	$\frac{\sqrt{2}}{3}a^3$
Height (vertex-to-vertex)	$\sqrt{2}\,a$
Inradius (r)	$\frac{\sqrt{6}}{6}a$
Circumradius (R)	$\frac{\sqrt{2}}{2}a$
Midradius (ρ)	$\frac{1}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$\mathrm{Spec}(A) = \{(4)^1, (0)^3, (-2)^2\}$

Laplacian Matrix

$$L = \begin{bmatrix} 4 & -1 & -1 & -1 & -1 & 0 \\ -1 & 4 & -1 & 0 & -1 & -1 \\ -1 & -1 & 4 & -1 & 0 & -1 \\ -1 & 0 & -1 & 4 & -1 & -1 \\ -1 & -1 & 0 & -1 & 4 & -1 \\ 0 & -1 & -1 & -1 & -1 & 4 \end{bmatrix}$$

$\mathrm{Spec}(L) = \{(6)^2, (4)^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{6})^2, (2)^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 4 axes and each face is a triangle -> 2 rotations per axis $(120^\circ, 240^\circ) = 4 \times 2 = 8$

Vertex to opposite vertex -> 3 axes and each vertex has degree 4 -> 3 rotations per axis $(90^\circ, 180^\circ, 270^\circ) = 3 \times 3 = 9$

Midpoint of opposite edges -> 6 axes and total rotations per axis -> 1 $(180^\circ) = 6 \times 1 = 6$

Total rotations = 1+8+9+6 = 24

$|Rot(Octahedron)| = 24$

All rotational symmetries of the octahedron correspond to permutations of the 4 pairs of opposite faces.

$Rot(Octahedron) \cong S_4$

Total reflections = Total rotations = 24

Total symmetries = total rotations + total reflections = 24+24 = 48

$|Sym(Octahedron)| = 48$

$Sym(Octahedron) \cong S_4 \times C_2$

Dodecahedron

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Geometric Properties

Let the edge length be a.

Define the golden ratio:

$$\phi = \frac{1+\sqrt{5}}{2}$$

Property	Formula / Value
Faces	12
Edges	30
Vertices	20
Face Type	Regular Pentagon
Face Interior Angle	108°
Dihedral Angle	$\cos^{-1}\left(-\frac{1}{\sqrt{5}}\right) \approx 116.57^\circ$
Surface Area	$3\sqrt{25+10\sqrt{5}}\,a^2$
Volume	$\frac{15+7\sqrt{5}}{4}a^3$
Pentagon Diagonal	ϕa
Inradius (r)	$\frac{a}{20}\sqrt{250+110\sqrt{5}}$
Circumradius (R)	$\frac{\sqrt{3}}{2}\phi a$
Midradius (ρ)	$\frac{1}{4}(3+\sqrt{5})a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix}
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\end{bmatrix}$$

$\mathrm{Spec}(A) = \{(3)^1, (\sqrt{5})^3, (1)^5, (0)^4, (-2)^4, (-\sqrt{5})^3\}$

Laplacian Matrix

$$L = \begin{bmatrix}
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 0 & -1 & 3
\end{bmatrix}$$

$\mathrm{Spec}(L) = \{(3+\sqrt{5})^3, (5)^4, (3)^4, (2)^5, (3-\sqrt{5})^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{3+\sqrt{5}})^3, (\sqrt{5})^4, (\sqrt{3})^4, (\sqrt{2})^5, (\sqrt{3-\sqrt{5}})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Centre of opposite faces -> 6 axes and each face is a pentagon -> 4 rotations per axis $(72^\circ, 144^\circ, 216^\circ, 288^\circ) = 6 \times 4 = 24$

Vertex to opposite vertex -> 10 axes and each vertex has degree 3 -> 2 rotations per axis $(120^\circ, 240^\circ) = 10 \times 2 = 20$

Midpoint of opposite edges -> 15 axes and total rotations per axis -> 1 $(180^\circ) = 15 \times 1 = 15$

Total rotations = 1+24+20+15 = 60

$|Rot(Dodecahedron)| = 60$

All rotational symmetries of the dodecahedron correspond to even permutations of the 5 inscribed cubes.

$Rot(Dodecahedron) \cong A_5$

Total reflections = Total rotations = 60

Total symmetries = total rotations + total reflections = 60+60 = 120

$|Sym(Dodecahedron)| = 120$

$Sym(Dodecahedron) \cong A_5 \times C_2$

Icosahedron

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Geometric Properties

Let the edge length be a.

Define the golden ratio:

$$\phi = \frac{1+\sqrt{5}}{2}$$

Property	Formula / Value
Faces	20
Edges	30
Vertices	12

Property	Formula / Value
Face Type	Equilateral Triangle
Face Interior Angle	60°
Dihedral Angle	$\cos^{-1}\left(-\frac{\sqrt{5}}{3}\right) \approx 138.19^\circ$
Surface Area	$5\sqrt{3}\,a^2$
Volume	$\frac{5(3+\sqrt{5})}{12}a^3$
Height (vertex-to-vertex)	$\phi\sqrt{5}\,a$
Inradius (r)	$\frac{\sqrt{3}}{12}(3+\sqrt{5})a$
Circumradius (R)	$\frac{1}{4}\sqrt{10+2\sqrt{5}}\,a$
Midradius (ρ)	$\frac{\phi^2}{2}a$

Spectral Properties

Adjacency Matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$\mathrm{Spec}(A) = \{(5)^1, (\sqrt{5})^3, (-1)^5, (-\sqrt{5})^3\}$

Laplacian Matrix

$$L = \begin{bmatrix} 5 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 5 & -1 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 5 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 5 & -1 & 0 & 0 & 0 & -1 & -1 & -1 & 0 \\ -1 & -1 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & -1 & -1 \\ 0 & -1 & -1 & 0 & 0 & -1 & 5 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 5 & -1 & -1 & 0 & -1 \\ -1 & 0 & -1 & -1 & 0 & 0 & 0 & -1 & 5 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & -1 & 5 & -1 & -1 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & -1 & 5 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & -1 & -1 & 5 \end{bmatrix}$$

$\mathrm{Spec}(L) = \{(5+\sqrt{5})^3, (6)^5, (5-\sqrt{5})^3, (0)^1\}$

Singular Values of Incidence Matrix

$\sigma(Q) = \{(\sqrt{5+\sqrt{5}})^3, (\sqrt{6})^5, (\sqrt{5-\sqrt{5}})^3, (0)^1\}$

Symmetries and Automorphisms

To count:

Identity = 1

Vertex to opposite vertex -> 6 axes and each vertex has degree (5) -> 4 rotations per axis $(72^\circ, 144^\circ, 216^\circ, 288^\circ) = 6 \times 4 = 24$

Centre of opposite faces -> 10 axes and each face is a triangle -> 2 rotations per axis $(120^\circ, 240^\circ) = 10 \times 2 = 20$

Midpoint of opposite edges -> 15 axes and total rotations per axis -> 1 $(180^\circ) = 15 \times 1 = 15$

Total rotations = 1+24+20+15 = 60

$|Rot(Icosahedron)| = 60$

All rotational symmetries of the icosahedron correspond to even permutations of the 5 cubes dual to the inscribed dodecahedral structure.

$Rot(Icosahedron) \cong A_5$

Total reflections = Total rotations = 60

Total symmetries = total rotations + total reflections = 60+60 = 120

$|Sym(Icosahedron)| = 120$

$Sym(Icosahedron) \cong A_5 \times C_2$

What makes a solid Platonic?

Definition

A solid is called **Platonic** if it is a **convex polyhedron with congruent regular polygonal faces** and possesses complete symmetry through **flag-transitivity**.

A *flag* consists of a vertex, an incident edge, and an incident face. A polyhedron is said to be **flag-transitive** if any flag can be mapped to any other flag by a symmetry (rotation or reflection) of the solid. This ensures that no face, edge, vertex, or local configuration is geometrically distinguished from another.

Consequently, a Platonic solid satisfies the following equivalent properties:

- All faces are congruent regular polygons.
- The same number of faces meet at every vertex.

- All edges have equal length.
- All vertices, edges, and faces are equivalent under symmetry.

These conditions formalize the idea of a **perfectly regular three-dimensional solid**. Convexity prevents self-intersections, regular faces ensure uniformity, and flag-transitivity guarantees complete geometric symmetry.

Relaxing the Conditions of Platonic Solids

Platonic solids arise from three fundamental conditions:

1. **Convexity** — the solid has no self-intersections or inward indentations.
2. **Regular polygonal faces** — every face is a congruent regular polygon.
3. **Flag-transitivity** — any flag (a vertex, an incident edge, and an incident face) can be mapped to any other flag by a symmetry of the solid.

These conditions are highly restrictive and uniquely determine the five Platonic solids. However, if one of the conditions is relaxed, new families of polyhedra emerge.

Removing Convexity

If convexity is removed while preserving regular polygonal faces and flag-transitivity, one obtains the **Kepler–Poinsot solids**. These are non-convex regular star polyhedra that retain complete symmetry but allow self-intersecting geometry.

Removing Flag-Transitivity

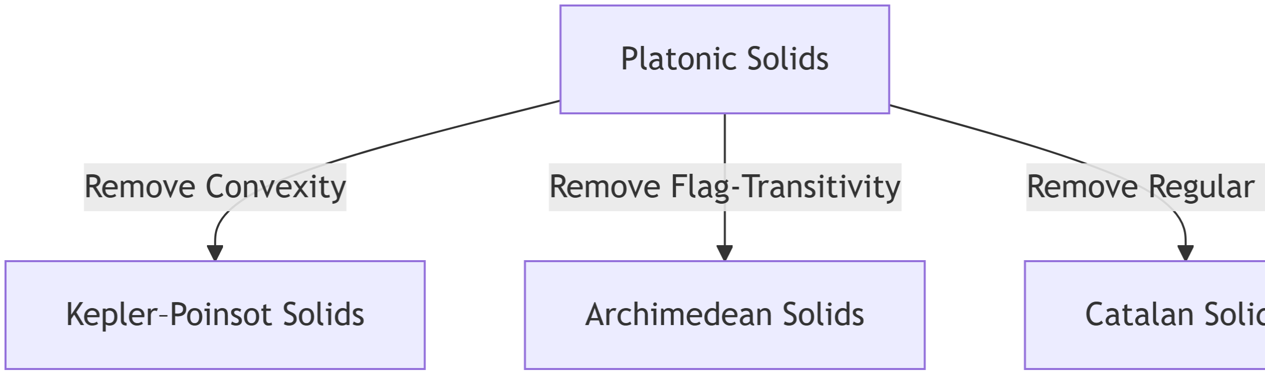
If flag-transitivity is removed while preserving convexity and regular polygonal faces, one obtains the **Archimedean solids**. These solids remain highly symmetric and vertex-transitive, but different face types may occur, preventing complete flag equivalence.

For example, the cuboctahedron contains both triangular and square faces, so a symmetry cannot map a flag involving a triangle to one involving a square.

Removing Regular Polygonal Faces

If regular polygonal faces are relaxed while preserving convexity and flag-transitivity, one obtains the **Catalan solids**, which are the duals of the Archimedean solids. These solids preserve strong symmetry, although their faces are no longer regular polygons.

Thus, the Platonic solids may be viewed as the **most restrictive and maximally symmetric class of convex polyhedra**, arising only when all three defining conditions are simultaneously satisfied.



Consequences of a Solid Being Platonic

Once a solid satisfies the defining conditions of Platonicity, several important geometric, combinatorial, and algebraic properties follow naturally.

Symmetry and Transitivity

Platonic solids exhibit the greatest degree of symmetry among convex polyhedra. Their symmetry groups act transitively on flags, implying that every incident vertex-edge-face triple can be mapped to any other through a symmetry of the solid.

Consequently, Platonic solids are:

- **vertex-transitive**, meaning every vertex is structurally identical,
- **edge-transitive**, meaning every edge occupies an equivalent position,
- **face-transitive**, meaning every face is symmetrically equivalent.

Thus, no vertex, edge, or face is distinguished from another, and every local region of the solid behaves identically.

This complete regularity distinguishes Platonic solids from other polyhedral families. For example:

- **Archimedean solids** are vertex-transitive but generally fail to be face-transitive, since they contain multiple types of regular polygonal faces.
- **Catalan solids** are face-transitive but fail to be vertex-transitive, since vertices may possess different local configurations.

The rotational symmetry groups of the Platonic solids are:

Platonic Solid	Rotational Symmetry Group
Tetrahedron	A_4
Cube	S_4

Platonic Solid	Rotational Symmetry Group
Octahedron	S_4
Dodecahedron	A_5
Icosahedron	A_5

These symmetry groups explain the strong uniformity observed in Platonic solids and contribute significantly to their structured spectral properties.

Regular Graph Structure

The graph associated with a Platonic solid is a **regular graph**.

If a Platonic solid has Schläfli symbol $\{p, q\}$, then every face is a p-gon and exactly q faces meet at each vertex. Hence, every vertex has degree q, making the adjacency graph q-regular.

For a q-regular graph,

$$A\mathbf{1} = q\mathbf{1}$$

where A is the adjacency matrix and $\mathbf{1}$ is the all-ones vector. Consequently, q is always an eigenvalue of the adjacency matrix.

Duality

Every Platonic solid possesses a Platonic dual.

The duality relationships are

$$\text{Cube} \leftrightarrow \text{Octahedron}$$

$$\text{Dodecahedron} \leftrightarrow \text{Icosahedron}$$

while the tetrahedron is **self-dual**.

In the dual polyhedron, vertices and faces interchange roles while preserving regularity.

Spectral Properties

The regularity and symmetry of Platonic solids strongly influence the spectra of their adjacency and Laplacian matrices.

Some immediate consequences include:

- the largest adjacency eigenvalue equals the vertex degree,

- repeated eigenvalues arising from symmetry,
- highly structured eigenspaces,
- spectral patterns reflecting geometric regularity.

These spectral properties provide insight into connectivity, combinatorial structure, and symmetry, making spectral graph theory an effective tool for studying Platonic solids.

Graph Invariants

Distance Matrices

The **distance matrix** of a graph describes the shortest path distance between every pair of vertices. For a graph G with vertices $ V(G)={v_1,v_2,…,v_n} $, the distance matrix D is defined as

$$D_{ij} = d(v_i, v_j),$$

where $d(v_i, v_j)$ denotes the length of the shortest path between vertices v_i and v_j.

Since $d(v_i, v_j) = d(v_j, v_i)$, the distance matrix is symmetric and therefore has real eigenvalues. The collection of eigenvalues of D is called the **distance spectrum**.

The **distance distribution** of a vertex describes the number of vertices located at each graph distance from it. For a vertex v, it is defined as

$$N_k(v) = |\{u \in V(G) : d(u, v) = k\}|,$$

where $N_k(v)$ represents the number of vertices at distance k from v.

Solid	Diameter	Distance Distribution	Distance Spectrum
Tetrahedron	1	$(1, 3)$	$\{3, -1^{(3)}\}$
Cube	3	$(1, 3, 3, 1)$	$\{12, 0^{(4)}, -4^{(3)}\}$
Octahedron	2	$(1, 4, 1)$	$\{6, 0^{(2)}, -2^{(3)}\}$
Dodecahedron	5	$(1, 3, 6, 6, 3, 1)$	$\{50, 0^{(9)}, (-7 + 3\sqrt{5})^{(3)}, -2^{(4)}, (-7 - 3\sqrt{5})^{(3)}\}$
Icosahedron	3	$(1, 5, 5, 1)$	$\{18, 0^{(5)}, (-3 + \sqrt{5})^{(3)}, (-3 - \sqrt{5})^{(3)}\}$

For the graphs of the Platonic solids, the distance distribution is identical for every vertex owing to their high degree of symmetry. Furthermore, the computed distance distributions exhibit **palindromic patterns**, reflecting the uniform and highly regular arrangement of vertices within these graphs.

Graph Energy

The **graph energy** is a spectral invariant defined using the eigenvalues of the adjacency matrix. If the adjacency spectrum of a graph G is

$$\mathrm{Spec}(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\},$$

then the graph energy is defined as

$$E(G) = \sum_{i=1}^{n} |\lambda_i|.$$

The graph energy measures the overall magnitude of the adjacency eigenvalues and provides insight into the structural properties of the graph.

Graph Energy of the Platonic Solid Graphs

Tetrahedron

$$E(G) = |3| + 3|-1| = 3 + 3 = 6$$

Cube

$$E(G) = |3| + 3|1| + 3|-1| + |-3| = 3 + 3 + 3 + 3 = 12$$

Octahedron

$$E(G) = |4| + 3|0| + 2|-2| = 4 + 0 + 4 = 8$$

Dodecahedron

$$\begin{aligned} E(G) &= |3| + 3|\sqrt{5}| + 5|1| + 4|0| + 4|-2| + 3|-\sqrt{5}| \\ &= 3 + 3\sqrt{5} + 5 + 0 + 8 + 3\sqrt{5} \\ &= 16 + 6\sqrt{5} \approx 29.4164 \end{aligned}$$

Icosahedron

$$\begin{aligned} E(G) &= |5| + 3|\sqrt{5}| + 5|-1| + 3|-\sqrt{5}| \\ &= 5 + 3\sqrt{5} + 5 + 3\sqrt{5} \\ &= 10 + 6\sqrt{5} \approx 23.4164 \end{aligned}$$

Spectral Gap

The **spectral gap** is another important spectral invariant that measures the separation between the two largest adjacency eigenvalues. For a regular graph, it is defined as

$$\Delta = \lambda_1 - \lambda_2,$$

where λ_1 and λ_2 denote the largest and second-largest adjacency eigenvalues, respectively.

A larger spectral gap generally indicates stronger connectivity, better expansion properties, and faster mixing of random walks on the graph.

Spectral Gap of the Platonic Solid Graphs

Tetrahedron

$$\Delta = 3 - (-1) = 4$$

Cube

$$\Delta = 3 - 1 = 2$$

Octahedron

$$\Delta = 4 - 0 = 4$$

Dodecahedron

$$\Delta = 3 - \sqrt{5} \approx 0.764$$

Icosahedron

$$\Delta = 5 - \sqrt{5} \approx 2.764$$

Eulerian and Hamiltonian

A graph is said to be **Eulerian** if it contains an Eulerian circuit, which is a closed trail that visits every edge of the graph exactly once.

A connected graph is Eulerian if and only if every vertex has even degree:

$$\deg(v) \equiv 0 \pmod 2$$

for every vertex v in the graph.

The Platonic solid graphs are all regular graphs, meaning that every vertex has the same degree. For a Platonic solid represented by the Schläfli symbol {p,q}, the value q represents the number of faces meeting at each vertex, and therefore also the degree of every vertex in the corresponding graph.

Hence, a Platonic solid graph is Eulerian if $ q *is even.* $ Therefore, among the Platonic solid graphs, only the octahedron graph is Eulerian.

A graph is called **Hamiltonian** if it contains a Hamiltonian cycle, which is a cycle that visits every vertex exactly once and returns to the starting vertex.

Unlike Eulerian graphs, Hamiltonian graphs are not characterized by vertex degrees alone. A necessary condition is that every vertex must have degree at least two:

$$\deg(v) \geq 2$$

for every vertex v.

All Platonic solid graphs satisfy this condition because their vertex degrees are at least three. Furthermore, due to their high symmetry, each Platonic solid graph contains a Hamiltonian cycle. Hence, all five Platonic solid graphs are Hamiltonian.

Determined by Spectrum (DS) graphs

Two graphs are said to be **cospectral** if they have the same spectrum; that is, they possess the same set of eigenvalues with the same multiplicities, even though their graph structures may be different.

For a graph G with adjacency matrix A, the adjacency spectrum is defined as

$$\operatorname{Spec}(G) = \{\lambda_1, \lambda_2, \dots, \lambda_n\},$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A.

A graph is said to be **determined by its spectrum (DS)** if its adjacency spectrum uniquely determines the graph up to isomorphism. In other words, if another graph H satisfies

$$\operatorname{Spec}(G) = \operatorname{Spec}(H),$$

then

$$G \cong H.$$

The DS property indicates that the adjacency spectrum contains sufficient information to uniquely recover the graph structure, despite being an algebraic representation of the graph.

The graphs of the five Platonic solids are all **determined by their adjacency spectra**. Consequently, no non-isomorphic graph shares the same adjacency spectrum as any of the Platonic solid graphs. This demonstrates that the spectral information of these highly symmetric graphs is sufficiently rich to uniquely characterize their underlying structure.

Thus, for the Platonic solid graphs, the adjacency spectrum serves as a **complete graph invariant**, uniquely identifying each graph up to isomorphism.

Eigenvalues and Multiplicity

The spectra of the Platonic solids exhibit several repeated eigenvalues. Such repetitions are not accidental; rather, they arise from the high degree of symmetry possessed by these graphs.

For a graph with little symmetry, one would generally expect most eigenvalues to be distinct. In contrast, the Platonic solids admit many rotational symmetries. These symmetries make certain directions within the graph indistinguishable from one another. As a result, the adjacency matrix cannot distinguish between these equivalent directions, causing multiple linearly independent eigenvectors to share the same eigenvalue.

The multiplicity of an eigenvalue is equal to the dimension of its eigenspace. An eigenvalue of multiplicity one corresponds to a unique direction, while a repeated eigenvalue corresponds to a collection of independent directions that are equivalent from the perspective of the graph. Rotational symmetries may transform one such direction into another without altering the associated eigenvalue.

This relationship can be formalized using the rotational symmetry group of the solid. Every rotational symmetry induces a permutation of the vertices and hence a permutation matrix P. Since rotations preserve adjacency, the adjacency matrix A commutes with every such permutation matrix:

$$AP = PA$$

Consequently, each eigenspace of (A) is preserved by the rotational symmetries. The symmetries act within the eigenspace, mixing its vectors while leaving the eigenvalue unchanged. Thus, the multiplicity of an eigenvalue reflects the presence of a family of symmetry-related directions encoded by the graph.

The repeated eigenvalues observed in the spectra of the Platonic solids may therefore be viewed as a manifestation of their underlying geometric symmetry.

Eigenspaces as Invariant Subspaces

For an eigenvalue λ, the corresponding eigenspace E_λ consists of all eigenvectors associated with λ, together with the zero vector. The dimension of E_λ is precisely the multiplicity of the eigenvalue.

The eigenspaces of the adjacency matrix possess an important symmetry property. Let P be the permutation matrix corresponding to a rotational symmetry of the solid. Since rotations preserve adjacency, the adjacency matrix A commutes with P:

$$AP = PA$$

Now let $v \in E_\lambda$. Then

$$Av = \lambda v$$

Applying P and using the commutativity relation gives

$$A(Pv) = P(Av) = P(\lambda v) = \lambda(Pv)$$

Thus Pv is also an eigenvector associated with λ. Consequently, every rotational symmetry maps vectors in E_λ back into E_λ.

This means that each eigenspace is an invariant subspace under the action of the rotational symmetry group. The symmetries of the solid do not move vectors between different eigenspaces; rather, they act entirely within each eigenspace. In this sense, every eigenspace forms a self-contained space on which the symmetries of the graph can act.

Geometrically, one may think of an eigenspace as a collection of equivalent directions preserved by the symmetries of the solid. Rotations may transform one vector in the eigenspace into another, but they never carry a vector outside the eigenspace. Thus each eigenspace captures a distinct aspect of the symmetry structure of the graph.

Geometric Interpretation of Eigenvalues, Multiplicities and Eigenspaces

The eigenvalues of the adjacency matrix provide information about the global structure of a graph. Associated with each eigenvalue is an eigenspace consisting of all eigenvectors corresponding to that eigenvalue. Together, the eigenspaces decompose the ambient vector space into simpler invariant components.

An eigenvector may be viewed as a pattern distributed across the vertices of the graph. The corresponding eigenvalue measures how this pattern interacts with the adjacency structure. Eigenvectors associated with the same eigenvalue exhibit the same adjacency behaviour and therefore belong to the same eigenspace.

The dimension of an eigenspace is called the multiplicity of the corresponding eigenvalue. Geometrically, the multiplicity determines the number of independent directions available within the eigenspace. An eigenvalue of multiplicity one corresponds to a single distinguished direction, whereas a repeated eigenvalue corresponds to a higher-dimensional space containing several independent directions. These directions are not isolated from one another; rather, they form a unified geometric object, namely the eigenspace.

For the Platonic solids, repeated eigenvalues occur frequently due to the high degree of symmetry present in the graphs. Rotational symmetries induce permutations of the vertices and hence permutation matrices that commute with the adjacency matrix. As a consequence, every eigenspace is preserved by the action of the rotational symmetry group.

If E_λ denotes the eigenspace corresponding to an eigenvalue λ, then any rotational symmetry maps vectors in E_λ to other vectors in the same eigenspace. Thus the symmetries act within the eigenspace rather than moving vectors between different eigenspaces. Each eigenspace therefore forms an invariant subspace under the action of the rotational symmetry group.

This invariance provides a geometric explanation for eigenvalue multiplicities. A repeated eigenvalue indicates the presence of several independent directions that are indistinguishable from the perspective of the graph. Rotational symmetries may transform one such direction into another while preserving the eigenvalue. Consequently, the eigenspace represents a collection of symmetry-related directions encoded by the graph.

The decomposition of the vector space into eigenspaces may therefore be viewed as a decomposition of the graph into independent symmetry-preserving components. Each eigenspace captures a particular manifestation of the graph's symmetry, while the multiplicities reveal the extent to which equivalent directions occur within that component. In this sense, the spectrum of a Platonic solid reflects not only its combinatorial structure but also its underlying geometric symmetry.

Spectral Decomposition of the Vector Space

Since the adjacency matrices of the Platonic solid graphs are symmetric, their eigenvectors corresponding to distinct eigenvalues are orthogonal. Consequently, the ambient vector space $\mathbb{R}^n$ admits a decomposition into the direct sum of eigenspaces:

$$\mathbb{R}^n = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$$

where E_{λ_i} denotes the eigenspace associated with the eigenvalue λ_i.

Geometrically, this decomposition partitions the vector space into mutually orthogonal components, each corresponding to a particular eigenvalue. Every vector in $\mathbb{R}^n$ can be uniquely expressed as a sum of vectors belonging to these eigenspaces. Thus the eigenspaces provide a natural coordinate-free decomposition of the space into simpler geometric pieces.

Within an eigenspace E_λ, the action of the adjacency matrix is particularly simple: every vector is merely scaled by the factor λ. The complexity of the adjacency matrix is therefore encoded in the collection of eigenspaces and their associated eigenvalues.

Furthermore, each eigenspace is invariant under the rotational symmetries of the graph. As a result, the decomposition of $\mathbb{R}^n$ into eigenspaces is simultaneously a decomposition into symmetry-preserving subspaces. The rotational symmetries act independently within each eigenspace, allowing different aspects of the graph's symmetry to be studied separately.

In this sense, the spectral decomposition breaks the graph into orthogonal geometric components, each carrying its own contribution to the overall symmetry structure of the Platonic solid.

Spectral Embedding

While spectral decomposition provides an algebraic splitting of $\mathbb{R}^n$ into orthogonal eigenspaces, this decomposition is not directly visible when n is large. Spectral embedding provides a way to visualize individual eigenspaces geometrically.

Let E_λ be an eigenspace of dimension m, with orthonormal basis

$$u_1, u_2, \dots, u_m$$

Each basis vector u_j assigns a real value to every vertex of the graph. Thus, for each vertex i, we obtain an m-dimensional coordinate vector:

$$(u_1(i), u_2(i), \dots, u_m(i))$$

This construction maps the vertices of the graph into $\mathbb{R}^m$, producing a geometric configuration associated with the eigenspace E_λ. This is referred to as the spectral embedding of the graph into that eigenspace.

In this way, each eigenspace gives rise to its own geometric realization of the vertex set. The full spectral decomposition

$$\mathbb{R}^n = \bigoplus_{\lambda} E_\lambda$$

therefore corresponds to multiple such geometric realizations, one for each eigenvalue.

Although these embeddings live in different coordinate systems, they all originate from the same underlying structure: the adjacency matrix of the graph. Each embedding highlights a different spectral component of the graph, revealing how the vertices are organized within that component.

For highly symmetric graphs such as the Platonic solids, these embeddings often reflect the underlying symmetry in a structured way, since each eigenspace is invariant under the action of the rotational symmetry group.

For plots, visit the following: Spectral Embeddings

Rotational Groups on geometric objects

Here, the alphabets a, b, c, d, e are arbitrary labels.

Tetrahedron

Permutation on 4 vertices

Rotation Type	Number of Axes	Angle	Permutation Form	Count
Identity	-	0°	$(a)(b)(c)(d)$	1
Vertex-face axis	4	120°	$(a)(b\,c\,d)$	4
Vertex-face axis	4	240°	$(a)(b\,c\,d)$	4
Edge axis	3	180°	$(a\,b)(c\,d)$	3

Total count $= 12 = |A_4|$

We see that the rotation permutations are only even.

$$Rot(Tetrahedron) \cong A_4$$

Cube

Permutation on 4 body diagonals

Rotation Type	Number of Axes	Angle	Permutation Form (on the 4 body diagonals)	Count
Identity	-	0°	$(a)(b)(c)(d)$	1
Face axis	3	90°	$(a\,b\,c\,d)$	3
Face axis	3	180°	$(a\,b)(c\,d)$	3
Face axis	3	270°	$(a\,b\,c\,d)$	3
Vertex axis	4	120°	$(a)(b\,c\,d)$	4

Rotation Type	Number of Axes	Angle	Permutation Form (on the 4 body diagonals)	Count
Vertex axis	4	240°	$(a)(b\,c\,d)$	4
Edge axis	6	180°	$(a)(b)(c\,d)$	6

Total count $= 24 = |S_4|$

$$Rot(Cube) \cong S_4$$

Octahedron

Permutation on 4 pairs of opposite faces

Rotation Type	Number of Axes	Angle	Permutation Form (on the 4 pairs of opposite faces)	Count
Identity	-	0°	$(a)(b)(c)(d)$	1
Vertex axis	3	90°	$(a\,b\,c\,d)$	3
Vertex axis	3	180°	$(a\,b)(c\,d)$	3
Vertex axis	3	270°	$(a\,b\,c\,d)$	3
Face axis	4	120°	$(a)(b\,c\,d)$	4
Face axis	4	240°	$(a)(b\,c\,d)$	4
Edge axis	6	180°	$(a)(b)(c\,d)$	6

Total count $= 24 = |S_4|$

$$Rot(Octahedron) \cong S_4$$

Dodecahedron

Permutation on 5 inscribed cubes

Rotation Type	Number of Axes	Angle	Permutation Form (on the 5 inscribed cubes)	Count
Identity	-	0°	$(a)(b)(c)(d)(e)$	1
Face axis	6	72°	$(a\,b\,c\,d\,e)$	6

Rotation Type	Number of Axes	Angle	Permutation Form (on the 5 inscribed cubes)	Count
Face axis	6	144°	$(a\,b\,c\,d\,e)$	6
Face axis	6	216°	$(a\,b\,c\,d\,e)$	6
Face axis	6	288°	$(a\,b\,c\,d\,e)$	6
Vertex axis	10	120°	$(a)(b\,c\,d)$	10
Vertex axis	10	240°	$(a)(b\,d\,c)$	10
Edge axis	15	180°	$(a\,b)(c\,d)(e)$	15

Total count $= 60 = |A_5|$

$$Rot(Dodecahedron) \cong A_5$$

Icosahedron

Permutation on 5 inscribed cubes

Rotation Type	Number of Axes	Angle	Permutation Form (on the 5 inscribed cubes)	Count
Identity	-	0°	$(a)(b)(c)(d)(e)$	1
Vertex axis	6	72°	$(a\,b\,c\,d\,e)$	6
Vertex axis	6	144°	$(a\,b\,c\,d\,e)$	6
Vertex axis	6	216°	$(a\,b\,c\,d\,e)$	6
Vertex axis	6	288°	$(a\,b\,c\,d\,e)$	6
Face axis	10	120°	$(a)(b\,c\,d)$	10
Face axis	10	240°	$(a)(b\,d\,c)$	10
Edge axis	15	180°	$(a\,b)(c\,d)(e)$	15

Total count $= 60 = |A_5|$

$$Rot(Icosahedron) \cong A_5$$

Rotational Groups on vertices

Full-symmetry Groups

Generators of Symmetry Groups

Hearing the Platonic Solids: Graph Sonification

Graph sonification is the process of representing graph structure through sound. Instead of studying a graph solely through visual diagrams, structural information is mapped to auditory parameters such as pitch, timbre, rhythm, and amplitude. This provides an alternative way of exploring graphs and can reveal patterns that may be difficult to identify visually.

Many graph-theoretic quantities can be sonified. Adjacency relations, shortest-path distances, and spectral properties all encode different aspects of structure and therefore produce different auditory representations. In this project, we focus on spectral graph sonification, where sound is generated from the eigenvalues and eigenspaces of the graph Laplacian.

The graph Laplacian captures important global properties of a graph. Its eigenvalues describe spectral modes of variation, while the dimensions of the corresponding eigenspaces reflect structural symmetries. Highly symmetric graphs often possess repeated eigenvalues, leading to higher-dimensional eigenspaces and characteristic spectral signatures.

These spectral quantities can be mapped to sound in a natural way. Eigenvalues determine pitch, while eigenvalue multiplicities influence sound texture through beating effects or harmonic enrichment. As a result, graphs with simple spectra tend to produce clear and isolated tones, whereas highly symmetric graphs often generate richer and more complex auditory textures.

The Platonic solids provide an ideal setting for spectral sonification. Their graphs are highly regular and possess large symmetry groups, which are reflected in their Laplacian spectra. By converting these spectra into sound, we obtain auditory representations of graph structure and symmetry, allowing the Platonic solids to be explored not only visually and algebraically, but also through listening.

The following page explores graph sonification in platonic solids: Graph Sonification